

Intelligent Network Topology Optimization Using LAM

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Outline

- 1 Phase 2: AI-Based Router Recommendation
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Phase 2: AI-Driven Router Selection

Objective

To transition from static mathematical formulas (CMBA) to a dynamic **Machine Learning approach** that predicts the optimal router for caching based on real-time network states.

- **Input:** Real-time network telemetry data.
- **Process:** Data logging via CSV → Model Training → Prediction.
- **Output:** The specific Router ID (R_{best}) that yields the highest probability of a cache hit and lowest latency.

Feature Engineering & Data Collection

To train our model, we simulated network traffic and logged the following key features into a dataset (.csv):

- ① **Cache Occupancy:** The current percentage of storage used in the router. High occupancy may indicate a need for eviction or bypassing.
- ② **Cache Hit Ratio (CHR):** Historical performance of the router in serving content.
- ③ **Centrality Measure Value (CMV):** The topological importance of the node (derived from Phase 1).
- ④ **Latency:** The time delay observed at the specific router node.

These features serve as the independent variables (X) for our predictive model.

Dataset Snapshot (Real Telemetry Data)

Live Simulation Logs

Below is a sample of the actual telemetry data collected during the evaluation phase. The model analyzes the **network metrics** to predict the **Best Router** (R_{best}) for the next hop.

Req ID	Occupancy	CHR	Predicted R_{best}
0	0.724	0.598	R19
1	0.825	0.587	R19
2	0.804	0.572	R19
3	0.786	0.558	R19
4	0.777	0.539	R12

Table: Fragment of `best_router.csv` showing key metrics and AI predictions.

Note: Occupancy and CHR are normalized values (0-1) used as input features.

Model Architecture: Random Forest Classifier

We selected an ensemble learning method to handle the non-linear relationships between network metrics.

Algorithm Selected: Random Forest Classifier

- **Why Random Forest?** It is robust against overfitting and handles high-dimensional data effectively by averaging multiple decision trees.

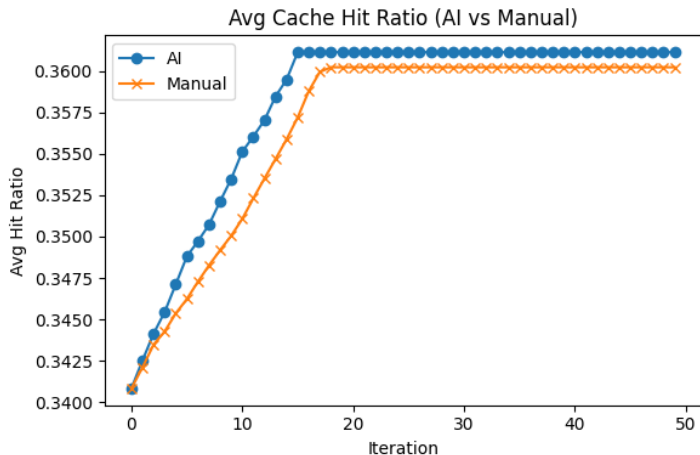
Hyperparameters Configured

- **n_estimators = 200:** We utilized 200 decision trees to ensure stability and reduce variance in predictions.
- **random_state = 42:** Fixed seed to ensure reproducibility of our evaluation results.
- **Target Variable:** The optimal Router ID for the specific content request.

- 1 **Simulation Phase:** The network simulator runs, calculating CMV, Latency, and Occupancy for every transaction.
- 2 **Data Logging:** All feature snapshots are appended to a training CSV file.
- 3 **Training:** The `RandomForestClassifier(n_estimators=200)` is trained on this historical data.
- 4 **Inference:** For new packets, the model analyzes current metrics and predicts the "Best Router."
- 5 **Action:** The packet is forwarded or cached at the predicted router, resulting in the performance improvements shown in the following graphs.

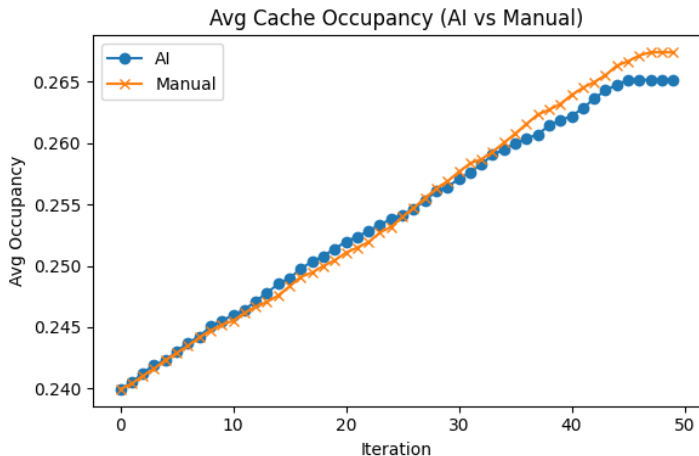
Performance Graph: Cache Hit Ratio (CHR)

This graph shows the improvement in Cache Hit Ratio over simulation iterations.



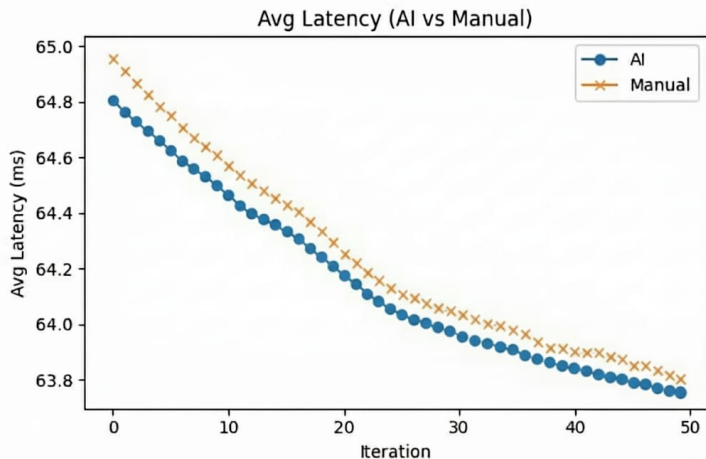
Performance Graph: Cache Occupancy

This graph illustrates the increase in the percentage of cache occupancy.



Performance Graph: Latency

This graph shows the average latency over number of iterations for AI recommended and Manual Method.



Expected Outcomes

- A more intelligent and adaptive network topology.
- Optimized cache placement for better resource utilization.
- Reduced content retrieval latency and redundancy in large-scale networks.

- **Simulator:** Custom Python Simulator
- **Algorithms:** Centrality Measures (CMBA), Reinforcement Learning (Q-Learning / Deep RL)
- **Languages:** Python
- **Evaluation Metrics:** Cache hit ratio, server hit reduction, hop reduction, latency, and throughput.

Our project aims to:

- Develop a scalable and intelligent topology optimization framework.
- Extend the centrality-based selection mechanism using a Large Action Model (LAM).
- Ultimately, create a more efficient Content-Centric Network for modern data demands.

Next Phase

Core Idea

To develop a Deep Reinforcement Learning (DRL) agent that dynamically optimizes caching and routing decisions in the network.

Objectives

- Implement and evaluate the baseline CMBA.
- Integrate a DRL agent to enhance decision-making.
- Evaluate overall performance improvement in terms of:
 - Cache Hit Ratio (CHR)
 - Hop Reduction Ratio (HRR)
 - Latency & Throughput



Kumari Nidhi Lal, Anoj Kumar (2017).

A Centrality-measures based Caching Scheme for Content-centric Networking (CCN).

Multimedia Tools and Applications.



AI-Driven Dynamic Cache Policy Selection in Information-Centric Networking (FYP Thesis 2024-25).

Thank You

Questions?